user access to the data might have averted this massacre by preventing the sale of firearms to the gunman.

These examples make a case for the importance of a distributed mechanism for timely information sharing. The sharing rules, conditions, and severity levels can be codified into a smart contract that will, in turn, enable recording of relevant metainformation in the distributed ledger of the blockchain.

1.4.3 Verifiable compliance

Let’s examine another area with numerous potential uses for the smart contract. Health care is a vast domain that has many requirements related to regulations and laws. Blockchain-based compliance, provenance, and governance can address many inefficiencies in this domain. Consider the Health Insurance Portability and Accountability Act (HIPAA), which is meant to protect the privacy and confidentiality of patient and other health data. Violations of this act by health care organizations or individuals may result in fines of anywhere from $1,000 to $250,000, so it’s in the best interest of everyone to keep track of how health care data is handled.

Compliance with HIPAA rules can be codified into smart contracts and recorded at the stakeholders’ blockchain node automatically, preventing any unwanted leaks of sensitive data. Businesses can ensure verifiable compliance. And blockchain provides a mechanism to demonstrate compliance to regulators.